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Welcome to another installment of CodeSport, which focuses on number theoretic algorithms. In particular, we will discuss the well-known 3-SUM problem, where given an array A of N numbers, we need to determine whether there exists a triple a, b and c that belongs to A , such that $a+b+c = 0$.

Thanks to all the readers who sent in their comments about the problems we discussed in the previous issue. Last

month's takeaway question was the birthday problem. As you enter a room with N people, what is the probability that there is someone in the room whose birthday is on the same day as yours? You can assume that there are no leap years and all years have only 365 days. Also assume that it's equally likely that the birthday is on any day of the year. None of the solutions I received from readers this month had the correct solution. Therefore, we will keep this problem open this month too. Here are some directions to help the reader arrive at the solution.

Let us number the N people in the room as $1, 2, 3, \dots$ up to N . We will use the loop index i to iterate over all people in the room and hence i can take values from $1, 2, 3, \dots$ up to N . Let ' bi ' be the day of the year in which the birthday of the person ' i ' falls. ' bi ' can take values from $1, 2, 3, \dots$ up to 365. If we assume that a person's birthday is equally likely to be any day of the year, the probability that a person's birthday falls on day ' D ' is given by $1/365$. In mathematical terms, we can say that probability of ' bi ' being equal to ' D ' is $1/365$, for any ' i ' from 1 to N and for any day from 1 to 365.

We can also assume that the birthdays of two people are independent events, in the sense that selecting ' D ' as the first person's birthday does not affect the outcome in the selection of the birthday of the second person. Since the events are independent, the probability that two

people ' i ' and ' j ' have the same birthday can be calculated by computing these two independent events. Since these are independent, the probability that the birthday of ' i ' is on day ' D ' and the birthday of ' j ' also is on day ' D ' is nothing but multiplication of these two individual probabilities. This can be written as: Probability of $(bi = D)$ and $(bj = D) = 1/365 * 1/365$.

Hence the probability that the birthdays of ' i ' and ' j ' both falling on any day can be obtained by summing this probability over all possible values for the days. Hence the probability of $(bi == bj) = \sum (1/365) * (1/365)$ over all days from 1 to 365.

Summing up $((1/365) * (1/365))$ over all days from 1 to 365, we find that the probability of $(bi == bj)$ is nothing but $1/365$. Let *Psame_birthday* be equal to the probability that two out of the N people have a matching birthday. We can use the probability we calculated for $(bi == bj)$ to show that $\sqrt{N}$ people need to be present in the room in order for *Psame_birthday* to be greater than 0.5. I leave it to the reader as an exercise. The hint to the solution is to consider the complementary event $E1$ of no two people in the room having the same birthday and then finding the probability of $(1 - P(E1))$ which gives the probability that at least two people in the room have the same birthday.

3-SUM problem—a naive algorithm

Let us get started with this month's well-known number theoretic problem known as the 3-SUM problem. Given a set of